

Hermetic Knowledge Isolation (HKI)

A runtime contract for enterprise agentic systems: every artifact belongs to one domain, every request runs in one active domain, and shared knowledge appears only through explicit publication rather than silent global fallback.

By Henok Ghebrechristos, PhD

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20 min

Full paper

5 min

Executive brief

6 visuals

System and threat
diagrams

HTML + PDF

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TYPICAL SYSTEMS

Scope drifts

Advisory scope weakens as requests move through rewriters, caches, graph traversal, and jobs.

HKI

Scope stays sealed

One active domain is selected at the edge and preserved through every downstream transformation.

SHARING

Publication is explicit

Shared knowledge crosses domains only through materialized copies, never through null or global fallback.

Choose Your Entry Point

Start with the version that matches your time and your goal. The full paper is the most rigorous artifact. The executive brief is the best entry point for leaders and broader audiences.

Executive Brief

A short decision-maker version of HKI focused on the problem, the contract, the leak paths it blocks, the architectural consequences, and the rollout path.

- ◆ Best for leaders, architects, and broad circulation
- ◆ Plain-language framing with the core visuals
- ◆ ~5 minute read

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Full Paper

The complete HKI writeup, including the formal model, threat analysis, related work, migration path, theory, scientific framing, and objections.

- ◆ Best for technical deep dives and design review
- ◆ Includes glossary, theorem sketch, and citations
- ◆ ~20 minute read

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The Picture

The contrast is straightforward: conventional "domain-aware" systems often treat scope as advisory, while HKI treats it as a sealed execution label.

Advisory scope

Sealed scope

Explicit publication

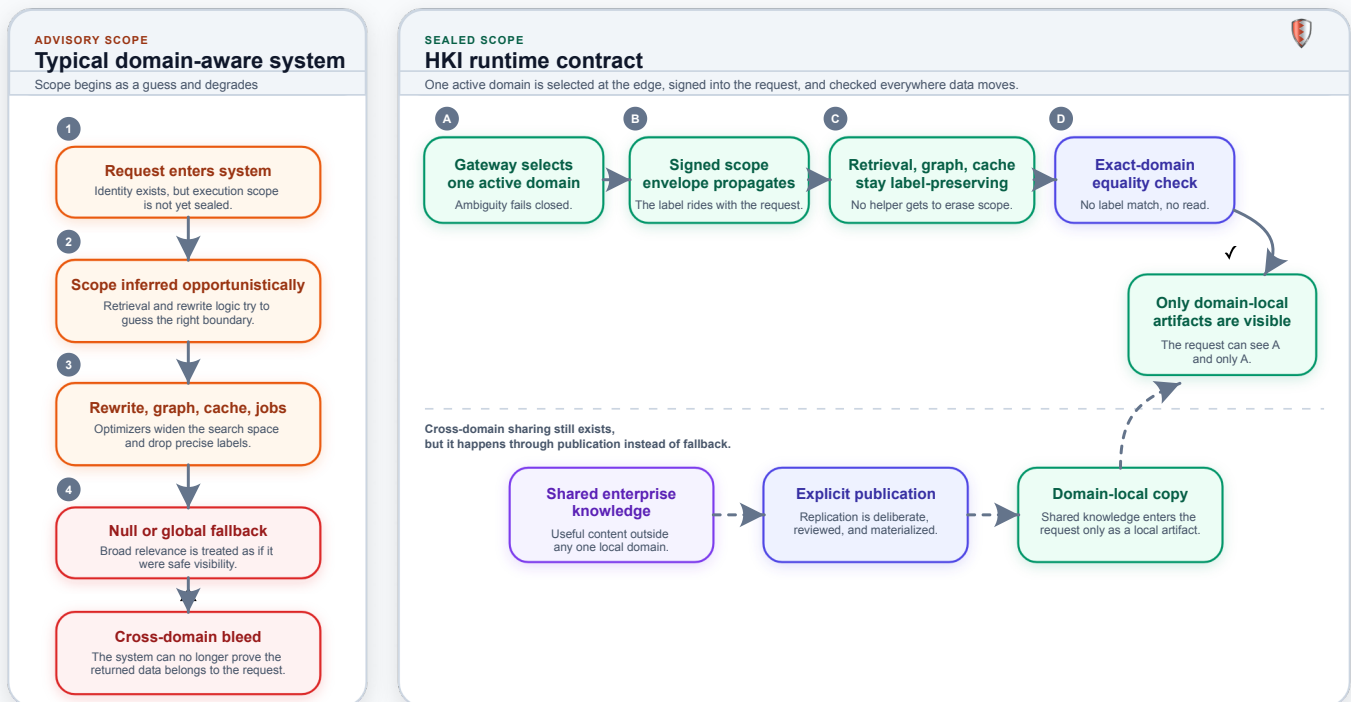
On the left, scope starts as an intention and erodes as the request passes through recall optimizers, graph traversal, semantic caches, and workflow shortcuts. On the right, HKI treats scope as part of execution itself.

That changes the system's posture. Instead of asking whether a result seems relevant enough to include, HKI asks whether the system is still operating inside the one domain selected for this request.

- ◆ Typical systems bind scope late and weakly.
- ◆ HKI binds scope early, signs it, and preserves it.
- ◆ Publication is the only bridge between domains.

The flow contrast at a glance

Conventional systems often weaken scope as they optimize.
HKI preserves one active domain from the gateway to every downstream decision.



The high-level contrast: scope drift versus sealed execution.

Three leak paths, all blocked

HKI does not rely on good intentions from helper subsystems.
Each attempted boundary violation is rejected against the active-domain contract.

LEAK PATH 1

Query-rewrite broadening

A retrieval helper expands a Domain A query so aggressively that it reads Domain B content.

1

A-labeled request enters recall

The system knows the active domain is A.
The question is whether helpers preserve it.

2

Query rewriter broadens toward B

The rewritten query becomes relevant across domains and attempts a B-labeled read.

3

HKI rejects: $\sigma(a)=B \neq \tau(r)=A$

Exact-domain equality stops the read
before data leaves B.

LEAK PATH 2

Derived-graph traversal

A graph edge appears convenient to follow, but it does not carry a valid scope label.

1

Traversal starts from valid node in A

The request begins on a legitimate A-owned artifact. The problem is what follows.

2

Unlabeled derived edge points into B

The graph wants to cross domains without proving that the edge itself belongs to A.

3

HKI rejects: derived edge lacks σ

No label-bearing edge, no traversal,
no cross-domain leak.

LEAK PATH 3

Cache contamination

A semantic cache finds a useful-looking result, but the key was not bound to the active domain.

1

Cache lookup starts from A request

The retriever asks for a fast answer on behalf of A. Correctness depends on the key.

2

Cache hit belongs to B

A naive key such as `org + query_hash` finds a result generated in another domain.

3

HKI rejects: key must bind domain

Cache state is partitioned by label,
so B cannot satisfy A.

All three leak paths fail closed

No helper subsystem overrules the active-domain contract — no cross-domain bleed occurs.

The concrete effect: three common leak paths all fail closed.

What You Get in the Full Package

Conceptual Model

TL;DR, executive summary, glossary, one-picture summary, and the four-part HKI thesis.

Implementation Guidance

Signed scope envelope, runtime/admin plane split, explicit publication model, threat analysis, and a five-phase rollout roadmap.

Defensibility

Novelty positioning, related work, theorem sketch, scientific framing, costs, limits, and anticipated objections.

Recommended Use

For a broad audience

Share the executive brief HTML or PDF.

For technical review

Use the full paper HTML for readability, or the PDF for a frozen artifact.

For design sessions

Open the full HTML and use the diagrams plus the threat and migration sections.

Feedback, critique, and counter-examples are welcome. The sharpest critique remains a concrete enterprise deployment in which null-domain runtime semantics are both safe and necessary.

Henok Ghebrechristos, 2026.